

Method for Creating an Authority File Database

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1 Purpose

The purpose of this process is to create a reference framework (authority file) that contains entities and relationships between them. This framework should be in the form of a graphical database and should contain necessary translations and alternate terminology for all entities.

2 System Requirements

Please note that these instructions are intended for a Windows operating system. The same procedure can be followed for MacOS, but the file locations and instructions intended specifically for File Explorer will be different.

Applications

Neo4j
Spreadsheet software (e.g. Google Sheets)

3 Methodology

Follow the instructions in the GitHub Repository section to ensure you have all of the required custom script downloads and extensions for appending the metadata.

3.1 OUTLINE THE NODES AND RELATIONSHIPS

System Requirements:

- *Google Sheets*

1. Open Google Sheets.
2. Create a table with columns for all of the entity keys you wish to include and one column for the relations. See Appendix A for an example.
3. Read each of the text pages and list the entities referenced in the text as individual rows. See Appendix A for an example.

For a full example of the spreadsheet, see the Entity Table file in the GitHub repository at [automation-of-methodologies-for-archival-processes](#) → [authority-file](#).

3.2 USE CYPHER TO CREATE THE DATABASE

System Requirements:

- *Text editor (e.g. Notepad, Visual Studio Code)*

1. Open a text editor (e.g. Notepad, Visual Studio Code).

2. Define all of the entities in the Google Sheet using MERGE statements like the one below.

```
MERGE (temp_id:Category {  
    property0: 'Property 0',  
    property1: 'Property 1',  
    property2: 'Property 2'  
})
```

- a. temp_id – the temporary id for the node used in this query.
 - b. Category – the category of the node (e.g. Dragon).
 - c. property# – the name of a property key of the node (e.g. label).
 - d. Property # – the information associated with the property# property key (e.g. Dragon).
3. Define all of the relationships in the Google Sheet using MERGE statements like the one below.

```
MERGE (temp_id0-[:RELATION]->(temp_id1))
```

- a. temp_id0 – the temporary id for the source node used in this query.
- b. temp_id1 – the temporary id for the target node used in this query.
- c. :RELATION – the defined relation between the two nodes (e.g. :TYPE_OF).

For a full example of Cypher code, see the authorityFile.txt file in automation-of-methodologies-for-archival-processes → authority-file. The [Cypher Manual](#) contains more information about MERGE statements and other commands you can use in Neo4j (Neo4j, n.d.).

3.3 RUN QUERIES IN NEO4J

System Requirements:

- Neo4j

1. Open [Neo4j](#)
 - a. Neo4j is a graphical database management and organization software (I. Neo4j, 2026).
 - b. *Create an account if you do not already have one.*
2. *If this is your first time using Neo4j,*
 - a. *Create an account.*
 - b. *Create an instance. All Neo4j databases run on instances.*

- i. *Select the Instances tab on the left, then select the blue “Create instance” button.*
 - ii. *Select the Free option, then name your instance and fill out the remaining information. It will take a few minutes for the software to finish creating the instance.*
3. Once your instance appears on the “Instances” page, select Open → Query.
4. Copy the Cypher code from your text editor and paste it into the text box at the top (“Start typing...”).
5. Select the blue run button. Your database should now be complete.

4 Installation Instructions

4.1 GITHUB REPOSITORY

The GitHub repository contains all of the necessary custom scripts required to follow this procedure.

1. Navigate to the GitHub repository [here](#).
2. Select the green “Code” button. Then select “Download ZIP.”
3. Navigate to your Downloads folder and extract the contents of the “automation-of-methodologies-for-archival-processes-main” folder.

5 Interactive Example

This example is the result of the Cypher code and entity list included in the shared GitHub repository. Follow the instructions below to view a completed database on Neo4j.

1. Navigate to workspace.neo4j.io.
2. Enter the following information.
 - a. Connection URL: `neo4j+s://13a7c7f8.databases.neo4j.io`
 - b. Database User: `13a7c7f8`
 - c. Password: `__BCM1nPsk2SLlaNwLZyov7FJm6QC9JkVOkmopT5tHY`
3. Select “Database Overview”. From here, you can see information about the database nodes, relationships, and property keys.
 - a. Nodes
 - i. The number next to the “Nodes” header indicates the number of nodes in the database.
 - ii. Each of the items below the header represents a different category of object.
 - iii. Double click on any of the items to see all the nodes in that category. The * will show all of the nodes. You may need to edit the provided code and remove “LIMIT #” to see the full database.
 - b. Relationships
 - i. The number next to the “Relationships” header indicates the total number of relationships in the database.
 - ii. Each of the items below the header represents a different type of relationship.
 - iii. Double click on any of the items to see all the relationships of that type. The * will show all of the nodes. You may need to edit the provided code and remove “LIMIT #” to see the full database.
 - c. Property Keys
 - i. The items below the header represent the different keys associated with each node.

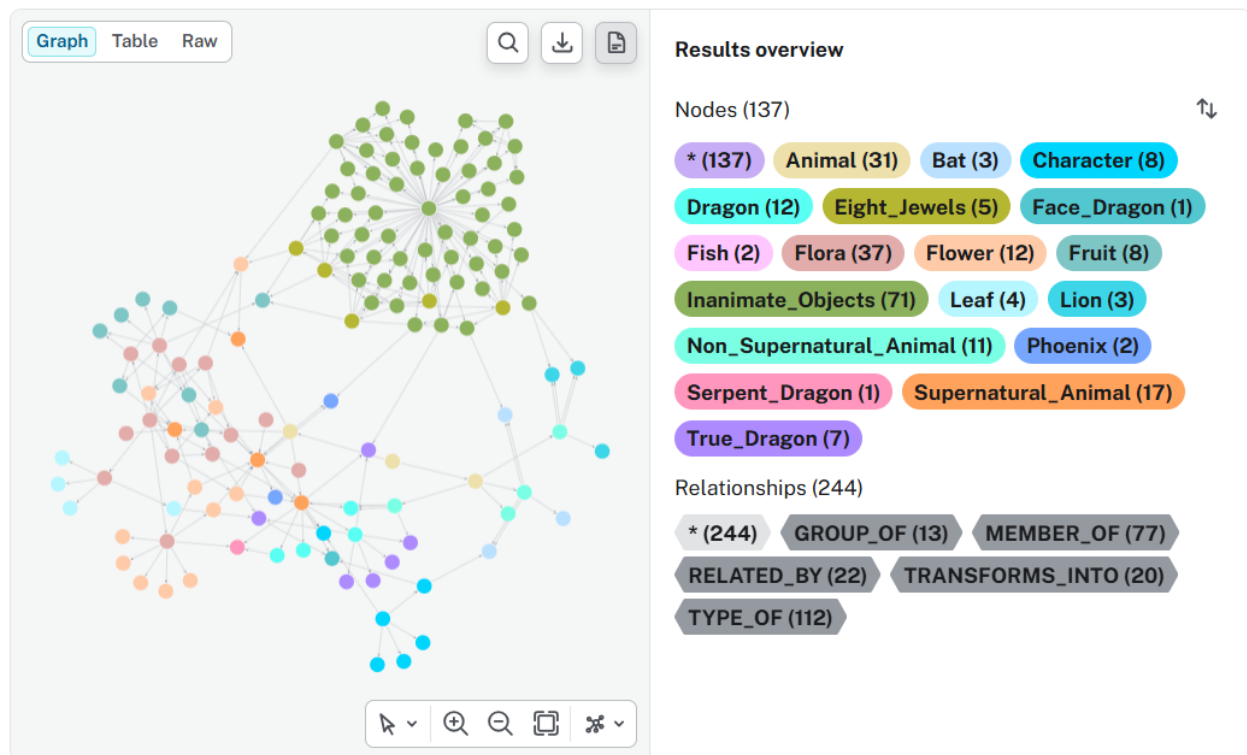


Figure 1: A screenshot of the full database created from the entities and relationships in chapters two through five of *L'Art à Hué*. The breakdown of the categories and relationships defined in the database is listed on the right.

Bibliography

- Neo4j. (n.d.). *Introduction - cypher manual* [Neo4j graph data platform]. Retrieved July 6, 2026, from <https://neo4j.com/docs/cypher-manual/25/introduction/>
- Neo4j, I. (2026, July 7). *Neo4j graph intelligence platform* [Graph database & analytics]. Retrieved July 6, 2026, from <https://neo4j.com/>

Appendix A: Entities and Relationships in *L'Art à Hué*

Columns in the Google Sheet

1. Entity Name (EN)
2. Entity Name (FR)
3. Entity Name (VN)
4. Entity Name (ZH)
5. Category
6. Relations

Entities and Relationships

1. Every entity should have an English (EN) and French (FR) translation.
2. Some will have a Vietnamese (VN) and/or a Chinese (ZH) translation.
3. The category refers to the chapter the entity was referenced in (e.g. Characters, Inanimate Objects, Flowers, etc.). The categories included in the *L'Art à Hué* authority file were:
 - a. Character
 - b. Inanimate Objects
 - i. Eight Jewels
 - c. Flora
 - i. Fruit
 - ii. Flower
 - iii. Leaf
 - d. Animal
 - i. Supernatural Animal
 1. Dragon
 - a. True Dragon
 1. Face Dragon
 - b. Serpent Dragon
 2. Pheonix
 - ii. Non-Supernatural Animal
 1. Bat
 2. Lion
 3. Fish
4. The relations to other objects should only be listed if they are part of any of the following five categories:

- a. The entity is a group of other objects.
 - i. Example: The Flaming Globe.
- b. The entity is a member of a group of objects.
 - i. Example: The Disc of the Moon, the Clouds, and the Flames are part of The Flaming Globe.
- c. The entity is a type of another entity.
 - i. Example: Dragon is a Supernatural Animal.
- d. The entity transforms into another entity.
 - i. Example: the Ribbons transform into the Flames.
- e. The entity is related to another entity via a third entity.
 - i. Example: the Dragon and the Fish are related by The Fish Changing into a Dragon.